

4.2 Vision Transformers

Image patches as tokens. The Week 3 block on pixels.

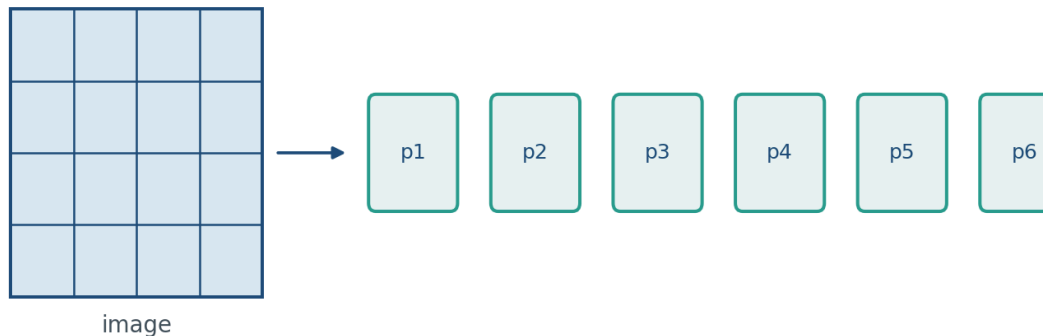
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A **vision transformer (ViT)** is the Week 3 block on **patches** instead of words. Once an image is a sequence, every LLM trick (attention, positions, [CLS], later CLIP) applies.

1. Patches as tokens

Split a $H \times W$ image into $P \times P$ patches, flatten each patch, map with a linear layer to width d .
 $N = HW / P^2$ tokens. Add a learned **[CLS]** if you classify, and add **positions** (row-column, or a 1-D index).

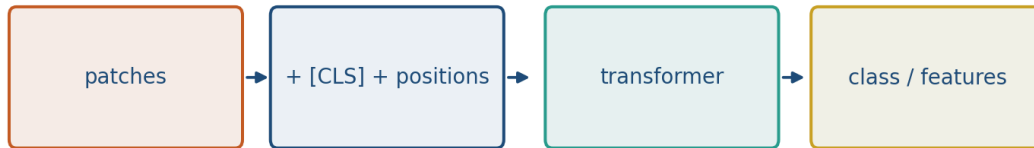
A ViT treats patches the way an LLM treats tokens



A CNN shares a small kernel and builds a hierarchy. A ViT sees global context in layer 1, at $O(N^2)$ cost. For a 224 image and $P = 16$, $N = 196$, which is a short paragraph.

2. The same stack

Vision transformer: same block as language, different tokens



Dosovitskiy et al. (2021): ViTs need more data than ResNets if trained from scratch; they shine with scale. In this course you will almost always **start from a pretrained** ViT (CLIP's image tower, or a Hugging Face checkpoint), not from random pixels.

Hybrid: a small CNN stem, then a transformer. Same idea.

3. Practice

1. Count tokens for a 384×384 image, patch 16.
2. Why add positions if attention already sees all patches?
3. A medical image dataset with 800 scans: from-scratch ViT or a pretrained tower? Why?